Morphological Classification of Galaxies in HST and JWST Images Using Classical DIP

Weekly Progress Report — Week 4

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ECE 501 – Digital Image Processing Course Project

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1 Week 4 Scope

Week 4 executes the next phase of the pipeline built on top of the finalized Week 3 restoration process (unsupervised Wiener deconvolution → bilateral denoising → Asinh stretch → JWST RGB composite). This phase encompasses segmenting and isolating discrete sources from the composite images, followed by computing quantitative Concentration–Asymmetry–Smoothness (CAS) parameters and geometric shape features for each segmented galaxy in Stephan's Quintet. All results described below reflect the corrected configurations after a dedicated code review.

2 Segmentation and Morphological Cleanup

2.1 Thresholding Configuration and Execution

Segmentation parameters were uniformly defined prior to thresholding. Key configurations include an adaptive threshold neighborhood size of 51 px, a sign-corrected threshold constant applied as $+5$, and a minimum connected-component area of 200 px.

Both targets were converted to 8-bit luminance using the mean of the RGB composite channels (for Stephan's Quintet) and the pre-processed frame (for Pillars of Creation). Thresholding was evaluated via two methods: global Otsu thresholding as a baseline and adaptive Gaussian thresholding. The adaptive method was selected as it successfully preserves faint extended structures that the single global Otsu cutoff inadvertently clips away.

2.2 Morphological Cleanup

A 5×5 elliptical structuring element was applied to the adaptive mask of each target in sequence. The process involved closing (2 iterations) to fill internal gaps, followed by opening (1 iteration) to remove isolated noise specks, and completed with binary hole-filling (`scipy.ndimage.binary_fill_holes`) to close any remaining interior voids.

3 Star–Galaxy Separation and Cutout Extraction

3.1 Classification Criteria

Connected components were labeled on the cleaned Stephan's Quintet mask, discarding components below 200 px in area. Each surviving blob was classified using a central-flux-fraction criterion: the flux inside the central 30% of the bounding box was compared to the total blob flux. An object was flagged as a star if its central fraction exceeded 0.35 and its total area was below 1500 px. These thresholds effectively target compact, centrally concentrated sources such as point sources and their diffraction spikes. All other blobs were kept as galaxy candidates.

3.2 Target Extraction

For Stephan's Quintet, each of the five catalogued members (NGC 7317, 7318A, 7318B, 7319, 7320) was matched to a surviving galaxy candidate in two stages. Initially, WCS astrometry (RA/Dec from the NED catalog projected to pixel coordinates) was used. If a usable WCS match failed within 500 px, a topological fallback assigned catalog IDs to the nearest unmatched blob under a fixed relative-position heuristic. Matched objects were cropped to their bounding box plus a 30 px padding margin and saved individually. For the Pillars of Creation, the three largest surviving structures by area were extracted using the same padding.

3.3 Quality Check and Limitations

The final labeled segmentation mask was overlaid on each composite with a distinct colour per object and matched bounding boxes drawn on top. Two known limitations were observed: NGC 7318A/B may merge into a single con-

nected component due to their close proximity, and faint tidal tails occasionally fell below the adaptive threshold, causing them to not be captured in the mask.

4 CAS Feature Extraction and Morphological Properties

Features were computed per extracted Stephan's Quintet cutout. The Asymmetry (A) and Smoothness (S) measurements both utilized a single fixed empty-sky reference patch drawn from the full composite (`raw_f200[100:300, 100:300]`) rather than a corner of each individual cutout to ensure consistent background noise estimation.

4.1 Concentration (C)

For each cutout, a flux-weighted photometric center was computed. Radial distance from that center was found for every unmasked pixel, and pixels were sorted by distance to build a cumulative curve of growth. Using the radii enclosing 20% (r_{20}) and 80% (r_{80}) of the total flux, the concentration index was computed as $C = 5 \cdot \log_{10}(r_{80}/r_{20})$.

4.2 Asymmetry (A) and Smoothness (S)

Each masked cutout was rotated 180° about its center and differenced against the original. The residual was normalized by total flux to yield a raw asymmetry. The same procedure on the fixed background patch provided a background asymmetry term (A_{bg}), yielding $A = \max(A_{\text{raw}} - A_{\text{bg}}, 0)$.

Similarly, cutouts were Gaussian-blurred ($\sigma = 3.0$ px) and differenced. Normalizing by total flux yielded raw smoothness. Using the background smoothness (S_{bg}) from the fixed sky patch, the final metric was $S = \max(S_{\text{raw}} - S_{\text{bg}}, 0)$.

4.3 Geometric Shape Moments

Two separate moment calculations were evaluated per cutout:

- **Binary moments:** Used to derive true pixel-count area and shape-only Hu invariants.
- **Intensity-weighted moments:** Used to determine the flux-weighted centroid, eccentricity, and ellipticity derived from the second central moments.

Perimeter was calculated from the external contour of the binary mask. The final metrics report both `area` (pixel count) and `total_flux` (intensity sum) as separate fields, correcting a previous conflation between the two.

5 Code Review Findings & Fixes Applied

Prior to generating final outputs, the processing algorithms underwent a code review to address several mathematical artifacts:

1. **Adaptive Threshold Sign Error:** The offset `C` parameter was effectively applying a negative shift instead of lowering the threshold. The sign was flipped, restoring the intended capability to preserve faint extended features.
2. **Background Region Estimation:** Asymmetry and Smoothness were previously normalizing against a 30×30 corner of each padded cutout. This introduced inconsistencies since cutout corners are not strictly empty sky. Both metrics now subtract a constant, verified empty-sky background noise patch taken from the global composite.
3. **Area Misclassification:** Original geometric extractions mislabeled flux-weighted `m00` values as physical pixel areas. The moments logic was bifurcated to compute both binary shape areas and separate intensity flux sums.

6 Generated Artifacts and Output Data

All geometric and CAS parameters (C, A, S, eccentricity, ellipticity, pixel area, total flux, perimeter, and seven log-scaled Hu moments) were assembled into `cas_features_week4.csv`. Literature sanity checks flagged any A or S components exceeding 0.8 as likely segmentation artifacts rather than genuine morphology. Output visuals generated include:

- `E1_threshold_comparison.png` (Otsu vs. adaptive thresholding)
- `E2_morphological_cleanup.png` (Closing / opening / hole-fill stages)
- `E3_star_galaxy_separation.png` (Star/galaxy overlay and scatter plots)
- `E4_sq_cutouts.png` (Gallery of matched cutouts)

- `E5_segmentation_overlay.png` (Final labeled mask and bounding boxes)
- `F1_concentration_cog.png` (Curve-of-growth plots)
- `F5_cas_distributions.png` (C/A/S bar charts with medians)

7 Planned Work — Week 5

Rule-based morphological classification (Elliptical / Spiral / Irregular decision tree) and benchmarking against Galaxy Zoo consensus labels are deferred to Week 5, per the project roadmap. The CAS and shape-moment table produced this week (`cas_features_week4.csv`) will act as the direct input for the classification decision logic.